

Yichao Jin, Ph.D.

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Specialization: Behavioral AI / Health Economics / Behavioral Economics / Discrete Choice Experiments

Education

- University of Texas at Dallas May 2026
Ph.D. in Public Policy and Political Economy
- Successfully Defended: April 14, 2026
- Dissertation: “Intertemporal Choice in Vaccination Timing: Evidence from a Discrete Choice Experiment in Wuhan”
- Committee: [Dohyeong Kim](#) (Chair), [Richard Scotch](#), [Soyoung Kwon](#), [Khai Chiong](#).
- Distinction: Dean of Graduate Education Dissertation Research Award (2025).
- University of Texas at Dallas May 2026
M.S. in Social Data Analytics and Research
- University of Queensland 2019
Master of Development Economics
- University of California, Riverside 2017
B.A. in Economics

Research Fields

Primary: Health Economics, Applied Econometrics, Behavioral Economics, Behavioral AI
Secondary: Quantitative Methods, Policy Evaluation, Technology Governance (AI in Health)

Working Papers

1. “Waiting Time as a Behavioral Barrier to Vaccination Uptake: Nonlinear Heterogeneity by Institutional Trust”
Job Market Paper — [Link to Full Draft](#)
 - Developed a structural Discrete Choice Experiment (DCE) framework to analyze the interaction between temporal barriers and institutional trust in vaccination behavior.
 - Identified nonlinear heterogeneity in waiting time sensitivity, providing a behavioral explanation for vaccine hesitancy.
2. “Empirical-Frontier Regularization for LLM Synthetic Agents: A Behavioral Digital Twin Framework”
Working Paper — [SSRN Link](#). Submitted to AI and Economics (AIE) Summer Conference 2026. Developed a 3-stage framework integrating DCE data and Structural Causal Modeling (SCM) to align generative agents with empirical choice frontiers, achieving a 93.8% reduction in frontier-deviation MSE.

3. **“Ideological Contamination in Policy Simulation: Separating Political Bias from Model Capabilities in Predicting Administrative Burden Effects”**
Working Paper — Investigates the disentanglement of inherent LLM ideological priors from structural predictive capabilities. Proposes a bias-correction methodology within the BDT framework to enhance the fidelity of administrative burden simulations.
4. **“The Ghost in the Machine: Algorithmic Rationality vs. Social Legitimacy in AI-Driven Public Health Governance”**
Working Paper — Analyzes the theoretical tension between automated efficiency and institutional trust using the AGIL framework and Rational Choice models.
5. **“The Price of Waiting: Evidence on Cash–Time Trade-Offs in Vaccination Time Discount”**
Manuscript in Preparation (Expected submission: August 2026) — Targeting *Social Science & Medicine*.

Research Experience & Affiliations

Doctoral Researcher, Spatial Health AI Research Partnership (SHARP), UT Dallas

- Leading interdisciplinary research on the intersection of artificial intelligence, geospatial analytics, and health policy.
- Managing large-scale datasets and developing computational simulations (R, Python) to evaluate pandemic preparedness.

Graduate Researcher, Behavioral Health Economics Laboratory, UT Dallas

- Conducted high-salience behavioral experiments (N=1,027) analyzing risk-weighted health decisions and institutional trust.

Teaching Experience

Teaching Assistant, School of Economic, Political and Policy Sciences, UT Dallas

- **Quantitative Methods for Policy Analysis (Graduate)**: Led Econometrics labs; specialized in R programming, regression analysis, and causal inference.
- **Health Economics & Public Policy (Undergraduate)**: Developed case studies on health market failures and administrative challenges.
- **Public Policy Analysis (Undergraduate)**: Mentored students in drafting evidence-based policy memos.

Conference Presentations

- **AI and Economics (AIE) Summer Conference 2026**: Empirical-Frontier Regularization for LLM Synthetic Agents.
- **APPAM Fall Research Conference 2025** (Seattle, WA): Estimating Time Discount Rate for Vaccination.
- **UT Dallas Graduate Research Symposium 2025**: Modeling Vaccination Decisions with Hyperbolic Discounting.

Honors & Awards

- **2025 Dean of Graduate Education Dissertation Research Award**, UT Dallas

- **2025** Betty & Gifford Johnson Graduate Travel Award, UT Dallas
- **2016** Omicron Delta Epsilon, International Honor Society for Economics

Technical Skills & Certifications

Econometrics: Mixed Logit (MXL), Latent Class Models (LCM), SEIR Modeling, Causal Inference.

Software: Stata (Advanced), R, Python, MATLAB, LaTeX, SQL, GIS.

Certifications: Stanford Artificial Intelligence Graduate Certificate; CITI Human Subjects Protection.

REFERENCES

Dohyeong Kim (Chair)

University of Texas at Dallas

Senior Associate Dean of Graduate Education

Robert E. Holmes Jr. Professor of Public Policy,

GIS & Social Data Analytics

Team Lead, SHARP

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Soyoung Kwon

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